



Trusted Aunti

Conversational Agentic RAG System

Trusted Aunti

A safety-aware Q&A proof of concept for adolescent girls in Kenya — grounded in vetted sources, routed by LangGraph, and evaluated beyond retrieval metrics.

Vetted sources

Privacy-first

Safeguarding-first



Trusted Aunti, refined around the outcome that matters

From a technically broad prototype to a simpler product focused on safe, grounded service access.

THE PRODUCT QUESTION

Can a trusted conversation help a girl move from uncertainty to a safe next action

- Grounded
- Private
- Safeguarded
- Useful



Alex Fulcher

Senior Director,
Tech & Product

Girl Effect

“The biggest impact for me is seeing a solution increase girls’ access to contraception in Ethiopia, helping them stay in education and have more control over their future.”

Paraphrased from our first interview conversation.

Framing: interview feedback + refined project scope. The downstream impact is the goal, not a claim established by this PoC.

I treated the feedback as a product experiment

The question was not “how do I make it smaller?” but “which parts actually earn their place?”

“The solution was overengineered in places. Show us a more refined version focused on delivery of the use case.”

PREVIOUS SHIPPED SYSTEM C

Broad, multi-model control graph

- 5 model roles** classification, evidence, generation, judging
- 2 LLM judges** evidence + output
- 4 content tracks** broader health/life-skills scope
- 12 / 51** unhelpful actions in the old benchmark

REFINED BUILD

Minimum justified complexity

- 1 hosted model** generation only
- Rules first** routing, safeguarding, validation
- 8 governed docs** contraception + sexual health + service access
- 3 / 52** deterministic unusable outcomes in the current run

What did not change: a universal safety floor · grounded factual answers · verified services · youth-facing conversation.

Note: the old and refined “unhelpful/unusable” instruments differ; the comparison is directional, not a precise percentage reduction.

A girl worried after unprotected sex, reaching HIV testing

Girl Effect’s behavioural journey translated into one conversation — from trusted discussion to a stated next action.



The product does not push her through stages. It removes the barrier in front of her and makes the next safe action easier.

* Self-reported next action, not verified HIV-test uptake. The PoC measures the conversation and service-access pathway, not downstream behaviour.

Journey: refined project README · Theory of Change framing: Girl Effect AI/ML whitepaper.

Refined architecture: complexity only where it earns its place

Safeguarding is a gate, not a route. Only generation uses a hosted LLM.



Resolve = understand the follow-up · Prepare = focus retrieval. Both are conditional.

Five tests, one product

Different failure modes need different tests — not one flattering chatbot score.



* Resonance here uses deterministic proxies. Warmth, trust and cultural authenticity still need human / youth review.

Previous System C vs refined build

A directional comparison of safety, usefulness, decision quality and model dependence.

WHAT WE MEASURED	PREVIOUS SYSTEM C	REFINED BUILD	WHAT CHANGED
Safety	0 unsafe actions 51-case benchmark	12/12 recall 12/12 precision	Safety retained
Usefulness	12 / 51 unhelpful actions	3 / 52 unusable outcomes	Fewer dead ends*
Decision quality	0.745 action accuracy	52 / 52 route accuracy	Rules replaced planner
Model dependence	3.53 LLM calls / turn	0.69–1.03 LLM calls / turn	≈3.4–5.1× fewer
REFINED BUILD — ADDITIONAL QUALITY CHECKS			
RETRIEVAL		GROUNDING	
Adequate@5 .960		24 / 24 cited	
		MULTI-TURN	
		forbidden 1 → 0	
		CONVERSATION	
		38 / 38 usable	

* Old = LLM-judged “unhelpful actions”; refined = deterministic blocked / no-evidence / provider-error outcomes. Instruments and datasets differ, so read this directionally.

Refined 3/52 = 1 validator block + 2 no-evidence + 0 provider errors; 8 correct out-of-scope declines are excluded.

The next proof comes from girls, clinicians and services

The prototype is ready. The next step is to validate it with the people and institutions around it.

1

Test with real girls

Run youth testing in Kenya to check trust, clarity, warmth and whether the conversation feels natural and useful.

2

Verify with health & safeguarding

SRH clinicians and safeguarding/policy reviewers validate HIV, contraception and referral content before wider use.

3

Measure real service access

Work with service partners to confirm contacts and measure whether intent leads to safe HIV testing or other SRH service uptake.



**The goal is not another model node —
it is safer conversations, stronger agency and real
access to care.**

Beyond the prototype: validate trust, validate content, validate that girls actually reach services.